

EE 210L: Digital Circuits and Logic Design Lab

Lab 4: Karnaugh Maps

Name: _____ Date: _____

Partner(s): _____ Section: _____

Record logic levels only, 0 or 1. Maps and logic diagrams stay on your Canvas pre-lab; carry only expressions and counts across.

Part 4: Carried Over from the Pre-Lab

Expressions only. Your maps and logic diagrams stay on the Canvas pre-lab.

Function	Expression
M, majority voter, from the map	
W, don't-cares used (hw 5.7d)	
W, every X forced to 0	
P, the form you will build (hw 5.34)	
P, the other minimal forms	
P, the eight codes that cannot occur	

Does your map answer for M agree with the expression you derived with the laws in Lab 3? Which method took fewer steps?

Part 5.1: Power Supply Setup

Measured power supply voltage: _____ V

Current limit setting: _____ mA

Part 5.2: Function W, What the Don't-Cares Bought You

Expression built W = _____

#	a	b	c	d	Predicted W	Measured W
1	0	0	0	0		
2	0	0	1	0		
3	0	0	1	1		
4	0	1	0	0		
5	0	1	0	1		
6	0	1	1	0		
7	0	1	1	1		
8	1	0	0	0		
9	1	0	1	0		
10	1	0	1	1		
11	1	1	0	1		
12	1	1	1	0		

The four rows below cannot occur in the system this function came from. Record what the circuit does anyway.

a	b	c	d	Measured W
0	0	0	1	
1	0	0	1	
1	1	0	0	
1	1	1	1	

Why were you free to ignore these four rows when you chose your groups?

Part 5.3: Function P, the Prime Detector

Expression built P = _____

#	Digit	a	b	c	d	Minterm	Predicted P	Measured P
1	0							
2	1							
3	2							
4	3							
5	4							
6	5							
7	6							
8	7							

Does the measured column match the specification: 1 for digits 1, 2, 3, 5, 7 and 0 for digits 0, 4, 6?

Part 5.4: Minimum Is Not Unique

Second form built P = _____

Digit	0	1	2	3	4	5	6	7
P from §5.3								
P from §5.4								

The two circuits agree on all eight valid digits. Do they also agree on the eight codes that cannot occur? Explain the result.

Part 5.5: Cost Comparison

Count 2-input gates. An n -input gate costs $n - 1$ of them. Count inverters separately.

Function	Version	Terms	AND	OR	NOT	Total gates	Packages
W	Don't-cares used						
W	X forced to 0						
P	Don't-cares used						
P	X forced to 0						

Would the all-zeros version of P fit the chips you brought? _____

Discussion and Conclusions

Summarize what you built and what it confirmed. Address at least: what the map does that the laws of Lab 2 cannot do, why a don't-care costs nothing to use, and what it means for two different expressions to be equally minimal.
